

Distinguishing Stop-Out from Attrition in Self-Paced Online Learning: A Computational Persistence Framework

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Abstract

Self-paced online learning environments, such as those at Athabasca University, allow students to register at any time, study for up to six months or longer, and pause their studies for work, family, health, or personal reasons. Most learning analytics systems treat prolonged inactivity as dropout. In self-paced programs, this assumption is demonstrably false: many students who stop out intend to return, and some continue learning offline while leaving few or no learning management system traces. This paper addresses the central research question: Can stop-out and attrition be distinguished computationally? We propose and formalize the Persistence-State Framework (PSF), which separates five states—Registered, Progressing, Stopped-Out, Withdrawn, and Completed—and introduces a latent Intent to Return variable, I_t . The PSF formalizes $\Pr(\text{Return} \mid \text{History})$ as a competing-risk problem in which Stopped-Out students may return to Progressing, convert to Withdrawn, or remain paused. We specify how the PSF can be implemented using Hidden Markov Models, Dynamic Bayesian Networks, survival analysis, and event history analysis, and we propose an operational validation strategy with explicit ground-truth definitions, label-generation procedures, and evaluation against traditional dropout models. To support empirical validation and institutional adoption, we also present the PSF Python library (psf-learning), an open-source reference implementation that provides the fitting, inference, validation, and visualization tools described in this paper. Data sources include Athabasca University course records: registration history, extension requests, tutor contacts, assignment submissions, and administrative withdrawals. This paper is a conceptual and computational framework paper; it does not report empirical model fitting or validation. The central contribution is a formal, testable, and computationally tractable account of temporary stop-out versus permanent attrition, with direct implications for early warning, intervention triage, and student support in self-paced distance education.

Keywords: stop-out, attrition, persistence, self-paced learning, hidden Markov models, survival analysis, event history analysis, learning analytics, intent to return, open-source software

1. Introduction

Student persistence and attrition are central concerns in online and distance education. In self-paced programs, however, persistence behaves differently than in cohort-based or term-based programs. Students may register at the beginning of any month, take up to six months or longer to complete a course, and are not bound by traditional term schedules (Athabasca University, n.d.). They may pause for weeks or months, engage with textbooks and other offline materials, and balance study against work and family commitments (Dron, 2007). As a result, the absence of visible activity in a learning management system (LMS) does not necessarily mean that a student has withdrawn. It may mean that the student has stopped out temporarily, intends to return, or is learning offline.

Most learning analytics systems collapse this complexity into a binary distinction: active versus inactive, or retained versus dropped out. This simplification is particularly problematic in online education, where students may temporarily interrupt their studies and later return, a behavior that has been described as stop-out rather than permanent attrition (Shefsky & Sutton, 2015). Inactivity thresholds are commonly used to flag students as dropout risks. This approach is administratively convenient but conceptually misleading. In self-paced distance education, a student who has not logged in for 30 days may be traveling for work, preparing for a proctored exam, reading a printed textbook, or waiting for an extension request to be processed. Treating all inactive students as dropouts overestimates attrition, misdirects interventions, and may alienate adult learners who fully intend to return.

Athabasca University (AU), Canada's open university, exemplifies the self-paced model. Completion rates for AU's learner-paced undergraduate courses have historically averaged around 63.6%, with non-completion rates substantially higher than those observed in grouped or paced study modes (Athabasca University, 2003). These figures underscore the need for models that distinguish temporary inactivity from permanent attrition. The practical difference is substantial: a student who is Stopped-Out may benefit from a reminder, a flexible extension, or a check-in from a

tutor; a student who is Withdrawn may need a different kind of support, such as re-registration advising or program re-entry planning.

This paper addresses that need by developing the **Persistence-State Framework (PSF)**, a focused conceptual and computational framework for persistence behavior in self-paced online learning. Unlike broader engagement models that attempt to represent behavioral, emotional, cognitive, social, and collaborative dimensions simultaneously, the PSF deliberately narrows its scope. It removes engagement taxonomy complexity and focuses entirely on persistence. The central research question is: **Can stop-out and attrition be distinguished computationally?** The central claim is that they can, provided that the model represents (a) persistence states rather than binary activity status, (b) a latent Intent to Return variable, and (c) the temporal history of registration, extension, tutor contact, and submission behavior.

It is important to state at the outset what this paper is and is not. This is a **conceptual and computational framework paper**. It proposes a formal theory, specifies mathematical models, defines operational procedures, and outlines an evaluation design. It does not report empirical model fitting, parameter estimation, or validation against institutional data. The PSF is designed to be implemented and tested in future empirical work. This positioning is deliberate: the contribution is the formalization of stop-out as a distinct computational problem, not the presentation of a fitted model.

The paper makes five contributions. First, it develops a formal theory of stop-out as a temporary state with an associated probability of return, $\Pr(\text{Return} \mid \text{History})$. Second, it introduces the latent Intent to Return variable I_t and specifies how it can be inferred from administrative and interaction data. Third, it provides computational specifications using Hidden Markov Models (HMMs), Dynamic Bayesian Networks (DBNs), survival analysis, and event history analysis. Fourth, it proposes an operational validation framework that includes ground-truth definitions, label-generation procedures, example trajectories, and a comparison against traditional dropout models. Fifth, it presents an **open-source reference implementation** — the PSF Python library — that operationalizes the framework and provides the tools needed for empirical validation on institutional data. The library is designed to reduce the barrier between the theoretical framework and its practical application, allowing researchers and institutional data scientists to fit the model, evaluate it against baselines, and generate publication-ready diagnostic figures without reimplementing the underlying algorithms.

The PSF is specified at the **course level**, although it can be extended to program-level persistence by treating courses as nested within programs, as discussed in Section 5.4. This scope decision reflects the fact that persistence decisions in self-paced learning are typically made course by course, and that course-level data are more readily available than program-level data in most institutional systems.

The remainder of the paper proceeds as follows. Section 2 reviews literature on persistence, attrition, and state-based modeling in distance education, and situates the PSF within established persistence theory and contemporary learning analytics. Section 3 presents the persistence-state theory, including operational definitions of attrition. Section 4 formalizes the computational model. Section 5 describes data sources and measurement. Section 6 presents the operational validation framework. Section 7 outlines evaluation and comparison against traditional dropout models. Section 8 discusses applications and integration with learning analytics systems. Section 9 considers limitations and future directions. Section 10 concludes.

2. Literature Review

2.1 Theoretical Foundations of Persistence

Persistence in higher education has been theorized extensively, though primarily in the context of traditional, cohort-based institutions. Tinto's (1975, 1993) Student Integration Model remains the most influential framework. Tinto argued that persistence is shaped by the fit between the student and the institution, mediated by academic and social integration. Students who feel academically and socially integrated are more likely to persist; those who do not are more likely to depart. Tinto later distinguished between different forms of departure, including transfer, temporary withdrawal, and permanent dropout, though his model was developed primarily for residential undergraduates.

Bean (1980) proposed an alternative model grounded in organizational behavior, arguing that student attrition resembles employee turnover. In Bean's model, persistence is influenced by background characteristics, academic

performance, social integration, and institutional commitment. Bean and Metzner (1985) extended this work to nontraditional students, who are more likely to be older, part-time, employed, and commuting. Their Nontraditional Undergraduate Student Attrition Model emphasized the role of external factors—work, family, and community commitments—in shaping persistence decisions. This emphasis is directly relevant to self-paced distance learners, who often fit the nontraditional profile.

Kember (1995) developed a model of student progress specifically for open and distance learning. Kember's model integrated Tinto's integration construct with the realities of distance education, including the absence of a physical campus and the centrality of learner autonomy. Kember emphasized that persistence in open learning is shaped by the interaction between student characteristics, institutional support, and the design of learning materials. Simpson (2013) extended this work by arguing that distance education institutions have historically failed to prioritize retention, and that persistence in distance learning requires proactive institutional intervention, particularly in the early stages of study.

These theoretical traditions converge on several points relevant to the PSF. First, persistence is not a binary outcome but a process that unfolds over time. Second, persistence is shaped by both institutional and external factors. Third, different forms of non-persistence—temporary pause, transfer, and permanent withdrawal—have different causes and different implications. Yet none of these theories explicitly formalizes the distinction between temporary stop-out and permanent attrition in a computational framework. This is the gap that the PSF addresses.

2.2 Persistence and Attrition in Self-Paced Distance Education

Research on open and distance education has shown that early engagement is a strong predictor of academic success. Subramaniam et al. (2024) found a positive correlation between early engagement and course completion in an open-distance program, with early-engaging learners demonstrating higher pass rates and lower instances of zero scores. Their findings also indicated that engagement patterns can serve as reliable proxies for self-regulated learning behaviors such as time management and proactive learning. This aligns with Zimmerman's (2002) model of self-regulated learning, which emphasizes forethought, performance, and self-reflection as cyclical processes that enable learners to manage their own study.

Rovai (2003) examined persistence in online programs and found that factors such as student characteristics, institutional support, and course design interact to shape retention. Park and Choi (2009) studied adult learners' decisions to drop out or persist in online learning and found that external factors—particularly family and work responsibilities—were more influential than institutional factors for many learners. Hart (2012) reviewed the literature on online persistence and concluded that no single factor explains dropout; instead, persistence is shaped by a complex interplay of personal, institutional, and contextual factors.

However, self-paced contexts complicate the measurement of persistence. Students may disengage from the LMS while continuing to read textbooks or complete assignments offline. The assumption that interaction frequency correlates with engagement level can therefore be misleading (Cocca & Weibelzahl, 2011). This has motivated calls for multi-dimensional models that integrate behavioral, cognitive, and affective indicators rather than relying solely on clickstream data. Yet for persistence specifically, the most urgent modeling problem is not the full dimensionality of engagement. It is the distinction between temporary stop-out and permanent withdrawal. Recent research in open online higher education suggests that stop-out and dropout often arise from different temporal and situational constraints, with stop-out students being more likely to re-engage once personal circumstances improve (Xavier et al., 2026).

2.3 The Limits of Binary Dropout Models

Traditional dropout models treat persistence as a binary outcome: a student either remains enrolled or drops out. In practice, dropout is often operationalized as a fixed period of inactivity—for example, no LMS login for 14, 21, or 30 days. This approach has three limitations.

First, it conflates observation with intention. A student who is inactive may intend to return, may be completing offline work, or may be waiting for an administrative process. Second, it ignores the temporal structure of persistence.

Dropout is not a single event; it is the result of a trajectory that may include pauses, returns, extensions, and near-withdrawals. Third, it produces biased estimates of attrition. If many inactive students eventually return, an inactivity-based model will generate false positives and overestimate the risk of withdrawal.

These limitations are especially severe in self-paced programs. In a cohort-based course, inactivity during a scheduled term may indeed predict dropout. In a self-paced course, inactivity is a normal part of the study cycle. A student may complete one assignment, pause for six weeks, and then return to complete the next. A model that labels this student as "dropped out" is not only inaccurate; it may trigger unnecessary or unwanted interventions.

2.4 State-Based and Temporal Models

State-based models offer a more appropriate framework for persistence in self-paced learning. Compartmental models adapted from epidemiology have been used to model student flows through educational programs. These models offer advantages for institutional planning and align with student lifecycle management. However, they typically operate at the aggregate level and do not capture individual-level latent states or the temporal dynamics of return.

Dynamic Bayesian Networks (DBNs) provide a natural formalism for individual-level temporal modeling. Murphy (2002) provided the foundational formalism for DBNs, and subsequent educational applications have demonstrated their superiority over static models for tracking engagement and performance over time (e.g., Zhang et al., 2024). The inference mechanism in Bayesian networks supports causal reasoning and dynamic belief updating, making them well-suited to adaptive learning environments (Wang & Han, 2016). Hidden Markov Models, a special case of DBNs, are particularly well-suited to persistence modeling because they represent a small number of latent states with transition probabilities and state-dependent observations (Rabiner, 1989).

Recent work in learning analytics has increasingly emphasized temporal and sequential modeling of learner states. Chen and Cui (2020) demonstrated the value of modeling student behavior as a time series, showing that temporal patterns captured through recurrent deep-learning architectures provided more accurate early prediction of academic outcomes than static approaches. This finding reinforces the importance of temporal representations when modeling persistence trajectories. Jovanović et al. (2017) used sequence analysis to unveil learning strategies in flipped classrooms, demonstrating that temporal patterns of activity are more informative than aggregate frequency counts. Matcha et al. (2019) reviewed the use of sequential and temporal methods in learning analytics and argued that state-based representations are essential for understanding self-regulated learning. Saint et al. (2020) combined analytic methods to understand patterns of self-regulated learning, showing that latent state models can capture the cyclical nature of forethought, performance, and reflection. These developments provide a methodological foundation for the PSF, which extends state-based temporal modeling to the specific problem of persistence in self-paced learning.

For self-paced learning specifically, deep learning approaches have been applied to dropout prediction. One study found that dropout rates among self-paced MOOC students exceed 90%, with over 50% participating for only one day (Early Prediction of MOOC Dropout, 2024). The proposed model accurately predicted over 98% of dropout cases and identified over 50% of engaged students. However, most existing models focus on prediction rather than explanation, and few explicitly represent the hierarchical structure of persistence or distinguish between temporary and permanent disengagement.

2.5 Research Gap

Although stop-out has been recognized as a distinct phenomenon in online student retention (Shefsky & Sutton, 2015), and empirical studies have examined stop-out behavior in higher education (Johnson, 2006), existing work has largely focused on descriptive analysis rather than computational state inference. Similarly, Xavier et al. (2026) demonstrated important differences between stop-out and dropout trajectories in open online higher education, but did not formalize these distinctions within a computational framework. Existing models of persistence in online learning rarely accommodate the specific conditions of self-paced study, including irregular sampling, offline learning, and the critical distinction between Stopped-Out and Withdrawn status. While state-based models exist, they have not been operationalized as individual-level computational models that infer Intent to Return from administrative and interaction data. Most predictive models focus on a single outcome—dropout—rather than representing the competing

risks of return, withdrawal, and completion. Finally, individual-level dynamic models and institutional-level compartmental models have developed largely in isolation.

The PSF addresses these gaps by developing a focused persistence-state framework. The PSF deliberately excludes the broader engagement taxonomy and concentrates on the states and transitions that matter for persistence: Registered, Progressing, Stopped-Out, Withdrawn, and Completed. It formalizes $\Pr(\text{Return} \mid \text{History})$ and introduces the latent Intent to Return variable I_t . It then specifies how the PSF can be implemented using HMMs, DBNs, survival analysis, and event history analysis, and proposes an operational validation strategy that compares it against traditional dropout models.

3. Theoretical Framework: The Persistence-State Framework

3.1 Persistence States

The PSF begins with five persistence states. These states are mutually exclusive and exhaustive for the purpose of persistence modeling, although a student may transition among them over time.

State	Label	Description
S_1	Registered	Enrolled in a course or program but not yet actively progressing.
S_2	Progressing	Actively completing assessments and advancing through course requirements.
S_3	Stopped-Out	Temporarily inactive with some probability of returning.
S_4	Withdrawn	Formally or effectively exited from studies with no intent to return.
S_5	Completed	Course or program requirements fulfilled.

The critical distinction is between S_3 (Stopped-Out) and S_4 (Withdrawn). Stopped-Out is a temporary interruption. The student may have paused for work, family, health, or financial reasons, but retains an intention to return. Withdrawn is a permanent exit. The student has formally withdrawn or has effectively abandoned studies with no intention to return. Conflating these states leads to overestimation of attrition and misdirected interventions.

This five-state model is a simplification of the nine-state Program Status dimension in earlier work. The simplification is intentional. By removing engagement taxonomy complexity, the PSF focuses entirely on persistence behavior. It treats "Progressing" as a single state rather than distinguishing among multiple engagement levels. It treats "Completed" as a terminal state for the course or program. The result is a parsimonious model that can be estimated from administrative data and evaluated against traditional dropout models.

3.2 Intent to Return

The central theoretical innovation of the PSF is the latent **Intent to Return** variable, I_t . At any time t , a student who is Stopped-Out may have $I_t = 1$ (intends to return) or $I_t = 0$ (does not intend to return). Intent to Return is not directly observed. It must be inferred from proxies such as:

- extension requests,
- tutor contacts,
- registration for future courses,
- communication with advising or support services,
- prior patterns of stop-out and return,
- timing of inactivity relative to course deadlines,
- submission of partially completed work.

Intent to Return is theoretically distinct from activity. This distinction is consistent with findings that students may temporarily withdraw because of situational constraints while retaining a strong intention to resume study once those constraints are resolved (Xavier et al., 2026). A student may be inactive but still intend to return. Another student may be active but already intend to withdraw. This distinction is what allows the PSF to separate temporary stop-out from permanent attrition.

3.3 Operational Definition of Attrition

A critical methodological issue in any study of stop-out and attrition is the ground-truth definition. Prior studies of stop-out behavior have highlighted the difficulty of distinguishing temporary interruption from permanent withdrawal using administrative records alone (Johnson, 2006). To address this challenge, the PSF proposes an operational definition of attrition that is independent of the latent-state model. If the model is used to infer withdrawal, and inferred withdrawal is then used as ground truth, the result is circular. To avoid this, the PSF proposes an operational definition of attrition that is independent of the model.

Definition 1 (Operational Attrition). A student is classified as **Withdrawn** if either:

- (a) a formal withdrawal record exists in the institutional administrative system, or
- (b) the student has not returned to active progress within a specified observation window W (e.g., 24 months) following the last recorded activity, and no formal extension or registration for a future course is active.

Definition 2 (Operational Stop-Out). The definition is consistent with prior conceptualizations of stop-out as a temporary interruption in educational participation with the potential for future re-engagement (Shefsky & Sutton, 2015; Johnson, 2006). A student is classified as **Stopped-Out** if:

- (a) the student has been inactive for a period exceeding a minimum threshold T_{pause} (e.g., 60 days), and
- (b) the student has not formally withdrawn, and
- (c) the student either returns to active progress within the observation window W or has an active extension, future registration, or documented intent to return.

These definitions have several important properties. First, they are operational: they can be applied to administrative records without reference to the latent state model. Second, they are time-bounded: the observation window W must be specified in advance. Third, they are falsifiable: a student classified as Stopped-Out at time t may later be reclassified as Withdrawn if they do not return within W . Fourth, they avoid circularity: the model is trained to predict these labels using data available before the classification date, not the labels themselves.

The choice of W is an institutional and theoretical decision. A shorter window (e.g., 12 months) produces a stricter definition of attrition and may classify some returning students as Withdrawn. A longer window (e.g., 36 months) produces a more permissive definition and may delay classification. In practice, we recommend reporting results under multiple values of W as a sensitivity analysis. For AU's learner-paced courses, where students may take up to six months or longer, a window of 24 months is a reasonable default.

3.4 Return Probability

The PSF formalizes the probability of return as:

$$\Pr(\text{Return} \mid \text{History}) = \Pr(Z_{t+k} = S_2 \mid Z_t = S_3 H_t)$$

where Z_t is the persistence state at time t , S_2 is Progressing, S_3 is Stopped-Out, and H_t is the history of observations and actions up to time t . In practice, this probability is computed over a horizon k , such as the next 30, 60, or 90 days.

The PSF also defines the competing risk of withdrawal:

$$\Pr(\text{Withdraw} \mid \text{History}) = \Pr(Z_{t+k} = S_4 \mid Z_t = S_3 H_t)$$

and the probability of remaining stopped out:

$$\Pr(\text{Remain Stopped-Out} \mid \text{History}) = 1 - \Pr(\text{Return}) - \Pr(\text{Withdraw})$$

These probabilities are the core outputs of the PSF. They allow institutions to distinguish between students who need a gentle reminder and students who need re-engagement support.

3.5 Propositions and Validity Considerations

The PSF yields several testable propositions.

Proposition 1. Under an inactivity-only model, Stopped-Out and Withdrawn students are observationally equivalent. A model that uses only activity recency cannot distinguish them.

Proposition 2. Intent to Return moderates the transition from Stopped-Out to Progressing or Withdrawn. Students with $I_t = 1$ have a higher hazard of return and a lower hazard of withdrawal than students with $I_t = 0$.

Proposition 3. Administrative and interaction signals—such as extension requests, tutor contacts, and future registrations—are informative proxies for Intent to Return.

Proposition 4. A persistence-state model that includes Intent to Return will reduce false positive attrition alerts compared with a traditional inactivity-threshold dropout model, without substantially reducing true positive withdrawal detection.

Proposition 2 requires careful justification because Intent to Return is latent. A reviewer may reasonably ask: how can a latent inferred variable be used to explain the phenomenon it is intended to predict? This is a question of construct validity, and it applies to many latent-variable models in the social sciences (e.g., motivation, self-efficacy, risk perception). We address it here on three levels.

Construct validity. Intent to Return is theoretically grounded in the persistence literature. Tinto (1993) distinguished between temporary and permanent departure and emphasized the role of institutional commitment and goal commitment in shaping persistence decisions. Bean and Metzner (1985) emphasized the role of external commitments—work, family, and community—in shaping nontraditional students' persistence. Kember (1995) emphasized the role of learner autonomy and institutional support in open learning. Intent to Return is a parsimonious construct that captures the student's commitment to continue, as distinct from their current activity level. It is not a convenience variable; it is a theoretical construct with antecedents and consequences specified in the persistence literature.

Measurement validity. Intent to Return is measured indirectly through proxies. This is common in latent-variable modeling: motivation, engagement, and self-regulation are also measured indirectly. The validity of the measurement model can be assessed through convergent and discriminant validity. Convergent validity would be supported if extension requests, tutor contacts, and future registrations all load positively on the latent variable. Discriminant validity would be supported if Intent to Return is empirically distinguishable from activity recency and academic progress. These are testable hypotheses, not assumptions.

Proxy validity. Not all proxies are equally valid. Extension requests and future registrations are likely to be stronger indicators than LMS logins, because they reflect deliberate action oriented toward continuation. Tutor contacts are intermediate: they may reflect either intent to return or intent to withdraw. The model should allow proxy-specific weights to be estimated rather than assuming equal validity. In the absence of strong proxies, the model should report higher uncertainty about Intent to Return rather than forcing a confident classification.

These considerations do not eliminate the challenge of latent-variable inference, but they frame it as a testable empirical question rather than an unexamined assumption.

4. Computational Model

4.1 Hidden Markov Model Formulation

The PSF can be implemented as a Hidden Markov Model (HMM). Let $Z_t \in \{S_1, S_2, S_3, S_4, S_5\}$ be the latent persistence state at time t . Let $I_t \in \{0, 1\}$ be the latent Intent to Return variable. Let Y_t be the vector of observations at time t , which may include activity counts, submission indicators, extension requests, tutor contacts, and administrative events.

The joint model is:

$$\Pr(Z_{1:T}, I_{1:T}, Y_{1:T}) = \Pr(Z_1) \Pr(I_1) \prod_{t=1}^{T-1} \Pr(Z_{t+1} | Z_t I_t) \Pr(I_{t+1} | I_t Z_t) \prod_{t=1}^T \Pr(Y_t | Z_t I_t)$$

The transition model for Z_t depends on the current state and Intent to Return. For a student in Progressing (S_2), the possible transitions are:

- remain Progressing,
- become Stopped-Out,
- become Withdrawn,
- become Completed.

For a student in Stopped-Out (S_3), the possible transitions are:

- return to Progressing,
- become Withdrawn,
- remain Stopped-Out.

The transition probabilities for Stopped-Out are:

$$\begin{aligned} \Pr(Z_{t+1} = S_2 | Z_t = S_3 I_t = 1) &= \lambda_{\text{return}}(t) \\ \Pr(Z_{t+1} = S_4 | Z_t = S_3 I_t = 1) &= \lambda_{\text{withdraw|intent}}(t) \\ \Pr(Z_{t+1} = S_3 | Z_t = S_3 I_t = 1) &= 1 - \lambda_{\text{return}}(t) - \lambda_{\text{withdraw|intent}}(t) \end{aligned}$$

For $I_t = 0$, the hazard of return is near zero and the hazard of withdrawal is higher:

$$\begin{aligned} \Pr(Z_{t+1} = S_2 | Z_t = S_3 I_t = 0) &\approx 0 \\ \Pr(Z_{t+1} = S_4 | Z_t = S_3 I_t = 0) &= \lambda_{\text{withdraw|no intent}}(t) \end{aligned}$$

The Intent to Return variable itself evolves over time. A simple model is:

$$\begin{aligned} \Pr(I_{t+1} = 1 | I_t = 1) &= \phi_1 \\ \Pr(I_{t+1} = 1 | I_t = 0) &= \phi_0 \end{aligned}$$

where ϕ_1 and ϕ_0 are persistence and recovery parameters. In practice, I_t is inferred from observations rather than directly observed.

4.2 Dynamic Bayesian Network Extension

The HMM can be extended to a Dynamic Bayesian Network (DBN) by incorporating covariates and hierarchical dependencies. Let X_t be a vector of covariates, such as course difficulty, time since registration, prior stop-out history, and tutor contact frequency. The transition model becomes:

$$\Pr(Z_{t+1}, I_t | Z_t, I_t, X_t)$$

and the observation model becomes:

$$\Pr(Y_t | Z_t, I_t, X_t)$$

A DBN allows the model to represent conditional dependencies among multiple data streams. For example, extension requests may depend on both the persistence state and the intent state. Tutor contacts may depend on the persistence state and the risk of withdrawal. Assignment submissions may depend on the progress state and the time since last activity.

The DBN formulation is particularly useful for handling irregularly sampled data. Self-paced learning generates observations at irregular intervals. A student may log in daily for a week, then disappear for a month. The DBN can propagate the latent state forward through periods of missing data and update beliefs when new observations arrive.

4.3 Survival and Event History Formulation

An alternative and complementary formulation uses survival analysis and event history analysis. In this formulation, the focus is on the timing of events: return, withdrawal, and completion. Let T be the time to an event, and let $K \in \{\text{return, withdrawal, completion}\}$ be the event type. The discrete-time hazard is:

$$h_k(t | H_t) = \Pr(T = t, K = k | T \geq t | H_t)$$

where H_t is the history up to time t . The overall survival function is:

$$S(t | H_t) = \exp \left(- \sum_{u=1}^t \sum_k h_k(u | H_u) \right)$$

The probability of return within a horizon K is:

$$\Pr(\text{Return by } t + K | H_t) = \sum_{u=t+1}^{t+K} S(u-1 | H_{u-1}) h_{\text{return}}(u | H_u)$$

This formulation is useful for two reasons. First, it directly models time-to-event, which is the natural outcome in self-paced learning. Second, it accommodates censoring. A student who is still Stopped-Out at the end of the observation window is censored, not treated as a withdrawal. This is a major advantage over binary dropout models, which must arbitrarily label censored students as dropouts or exclude them.

4.4 Inference and Learning

Parameters can be estimated using Expectation-Maximization (EM) for HMMs, Baum-Welch for discrete-state models, or Markov Chain Monte Carlo (MCMC) for DBNs. For real-time applications, particle filtering provides online state estimation:

1. **Predict:** Sample $Z_{t+1}^{(i)} \sim \Pr(Z_{t+1}, l, Z_t^{(i)} I_t^{(i)} X_t)$.
2. **Update:** Weight particles by $w^{(i)} \propto \Pr(Y_{t+1}, l, Z_{t+1}^{(i)} I_{t+1}^{(i)} X_{t+1})$.
3. **Resample:** Resample particles according to weights.

This yields the filtering distribution $\Pr(Z_t, I_t | Y_{1:t})$, from which practitioners can compute the probability of return, the probability of withdrawal, and the expected time to return.

For survival models, parameters can be estimated using partial likelihood for Cox models or discrete-time logistic regression for discrete-time hazards. Competing risks can be modeled using cause-specific hazards or subdistribution hazards.

4.5 Reference Implementation: The PSF Python Library

To support empirical validation and institutional adoption, the framework described in Sections 3 and 4 has been implemented as an open-source Python library, `psf-learning`. The library provides a complete, tested, and documented reference implementation of the computational model, the inference algorithms, the operational validation framework, and the diagnostic tools described throughout this paper. It is intended for three audiences: (a) researchers who wish to replicate or extend the PSF on institutional data, (b) institutional data scientists who want to integrate the framework into learning analytics pipelines, and (c) reviewers and methodologists who want to inspect the mechanics of the model directly.

The library is distributed under the MIT License and is available at https://github.com/hongxueharriswang/psf_lib. It requires Python 3.9 or later and depends only on NumPy, SciPy, scikit-learn, and pandas for its core functionality. Visualization utilities depend on matplotlib but are optional. This minimal dependency footprint is deliberate: it allows the library to be deployed in a wide range of institutional environments, including those with restricted software installation policies, without requiring specialized machine learning frameworks.

4.5.1 Design Principles

Three design principles guided the implementation.

Faithfulness to the formal model. Every component of the library corresponds directly to a mathematical object in Sections 3 and 4. The `PSFModel` class encapsulates the joint-state HMM; the `baum_welch` function implements the EM algorithm described in Section 4.4; the `forward_backward`, `viterbi`, and `particle_filter` functions implement the three inference procedures; and the `CompetingRiskSurvival` class implements the survival formulation in Section 4.3. Users who want to inspect or modify the model can trace each line of code back to its mathematical counterpart.

Separation of concerns. The library separates five concerns that are often entangled in learning analytics code: (a) data preparation (`features.py`, `labels.py`), (b) model fitting (`learning.py`), (c) inference (`inference.py`), (d) evaluation (`evaluation.py`, `baselines.py`), and (e) visualization (`viz.py`). This separation makes the pipeline reproducible and auditable, and it allows users to replace individual components without rewriting the rest.

Reproducibility by default. The library uses explicit random seeds, records the full EM log-likelihood history on each fitted model, and provides a synthetic simulation module (`simulation.py`) that generates cohorts under a known true model. This makes the framework's inference machinery fully verifiable: a user can fit a model to synthetic data and compare the fitted parameters to the true ones, as demonstrated in the reference simulation study described below.

4.5.2 Core Capabilities

The library provides the following capabilities.

Model fitting. The `baum_welch` function implements the EM algorithm described in Section 4.4. It respects the structural constraints of the joint state space (the two S_3 variants, the absorbing S_4 and S_5 states, and the forbidden transitions), handles missing data through explicit per-feature masks, and records the per-iteration log-likelihood for convergence diagnostics. The library recommends multi-restart fitting, in which the EM algorithm is run from several random initializations and the fit with the highest training log-likelihood is retained. In the reference simulation described below, multi-restart fitting raised the fitted model's AUC from 0.685 with a single restart to 0.998 with five restarts — a result that underscores the importance of avoiding local optima when the likelihood surface has multiple basins.

Inference. The `forward_backward` function computes the joint-state posterior $\Pr(Z_t, I_t \mid y_{1:t})$ and the smoothing posterior $\Pr(Z_t, I_t \mid y_{1:T})$. The `viterbi` function computes the most likely joint-state sequence. The `particle_filter` function provides online state estimation for real-time applications. The `intent_posterior` helper extracts $\Pr(I_t = 1 \mid \text{history})$ from the joint posterior, and the `return_probabilities` and `withdrawal_probabilities` functions compute the horizon-specific probabilities described in Section 3.4.

Operational label generation. The `generate_operational_labels` function implements Definitions 1 and 2 of Section 3.3. It accepts a list of `StudentRecord` objects and produces integer labels (0 = Completed, 1 = Withdrawn, 2 = Stopped-Out, 3 = Censored) that are independent of the model. This is the mechanism by which the framework avoids the circularity problem described in Section 3.3.

Baselines. The library includes three baseline models for comparison: `InactivityThresholdClassifier` (the standard operational definition of dropout), `LogisticDropoutModel` (a static binary classifier on mean features), and `SurvivalWithoutIntent` (the survival model of Section 4.3 with the Intent to Return variable removed). These baselines allow researchers to test hypotheses H1 through H4 of Section 7.3 directly.

Evaluation. The `evaluate_predictions` function computes AUC, PR-AUC, Brier score, precision, recall, F1, and false positive rate at a user-specified threshold. The `evaluate_at_positive_rate` function performs matched-positive-rate evaluation, in which all models are required to flag the same fraction of students — an operationally meaningful comparison that mirrors the constraint that advisors can contact only a limited number of students per week. The `concordance_index` and `time_dependent_auc` functions support survival-model evaluation.

Visualization. The `viz` module provides thirteen plotting functions that generate publication-ready diagnostic and comparison figures. These include EM convergence plots, transition-matrix heatmaps, emission-mean comparisons, individual state-trajectory plots, intent-trajectory plots, ROC and precision-recall curves, confusion matrices, and matched-rate metric comparisons. All figures are written at 150 DPI and can be saved as vector graphics for journal submission.

4.5.3 Reference Simulation Study

The library includes an end-to-end simulation study (`examples/01_simulation_study.py`) that verifies the inference machinery on synthetic data. The study builds a "true" PSF model with known parameters, simulates a cohort of 400 students over 36 monthly time steps with 20% missing data, fits a new model from scratch via multi-restart Baum-Welch, and evaluates both parameter recovery and predictive performance.

Parameter recovery. After aligning the fitted states to the true states using the Hungarian algorithm, the fitted transition matrix closely matches the true one. The diagonal entries of the persistence states are recovered to within 0.03, the $S_3(I=0) \rightarrow S_4$ transition is recovered to within 0.01, and the $S_3(I=1) \rightarrow S_2$ return transition is recovered to within 0.02. Emission means for S_1 , S_2 , $S_3(I=0)$, $S_3(I=1)$, and S_4 recover cleanly; S_5 shows a mild bias toward the population average, a small-sample effect from the absorbing state.

Latent-state recovery. Joint-state accuracy (the fraction of timesteps where the posterior-argmax state matches the true state) is 0.859, well above the 0.167 chance level for six states. Viterbi accuracy is 0.912.

Intent recovery. For students who pass through the Stopped-Out state, the Intent-to-Return posterior tracks the true intent label with an AUC above 0.95. This is the central verification of the framework's mechanism: the latent variable, which the manuscript claims is inferable from observable proxies, is in fact recovered with high accuracy.

Predictive performance. At the default 0.5 threshold, the PSF achieves AUC 0.998, PR-AUC 0.999, and F1 0.974. At the matched positive rate of 59.8% (equal to withdrawal prevalence in the sample), the PSF achieves precision 0.987, compared to 0.895 for the inactivity threshold, 0.862 for logistic regression, and 0.828 for survival without intent. At the operational capacity of 239 students, the PSF flags 236 true withdrawals — 22 fewer false positives than the inactivity threshold.

These simulation results verify that the inference machinery is correct and that the framework outperforms standard baselines when the data are generated by the model. They do not establish that the PSF is a better model of real self-paced learners; that question requires the empirical validation described in Section 7 and discussed further in Section 9.3.

4.5.4 Extensibility

The library is designed to be extended. Three extension points are anticipated.

Alternative emission families. The default emission is Gaussian. For count data (e.g., number of forum posts) or binary data (e.g., submitted assignment or not), users can subclass `PSFModel` and override `log_emission_prob` to implement Poisson, negative binomial, or Bernoulli emissions.

Covariate-aware transitions. The DBN extension described in Section 4.2 can be implemented by extending `PSFModel` with a covariate-dependent transition model. The library's structural-constraint machinery (the `ALLOWED_TRANSITIONS` mask and the `_refresh_A` method) handles the parameterization automatically once the covariate model is specified.

Deep extensions. The linear transition and observation models can be replaced with recurrent neural networks by wrapping the library in a differentiable probabilistic programming framework such as Pyro or TensorFlow Probability. The joint-state structure and the structural constraints remain unchanged; only the parameterizations of the transition and emission functions are replaced.

The library documentation (available at https://hongxueharriswang.github.io/psf_lib/) includes a user guide covering data preparation, model fitting, validation workflow, and diagnostic interpretation, along with a full API reference and a set of worked examples.

5. Data and Measurement

5.1 Data Sources

The PSF is designed for data from self-paced institutions such as Athabasca University. The primary data sources are:

Data source	Examples	Relevance
Registration history	Course registrations, start dates, program enrollment	Defines entry into Registered and Progressing states.

Extension requests	Requests for course extensions, approvals, denials	Proxy for Intent to Return.
Tutor contacts	Emails, phone calls, meeting requests	Proxy for Intent to Return and risk of withdrawal.
Assignment submissions	Submission times, scores, partial completions	Defines Progressing and Completed states.
Administrative withdrawals	Formal withdrawal records	Defines Withdrawn state.
LMS activity	Logins, page views, video plays	Auxiliary indicator of activity, not a definitive measure of persistence.
Completion records	Course completion, program graduation	Defines Completed state.

The model can be estimated on a weekly or monthly time grid. A weekly grid is appropriate for early warning; a monthly grid is appropriate for program evaluation.

5.2 Measurement of States

The latent states are partially observed. Registered is observed at registration. Progressing is inferred from assignment submissions and LMS activity. Stopped-Out is inferred from a period of inactivity combined with the absence of a formal withdrawal. Withdrawn is observed from administrative records or inferred from a long period of inactivity with no intent-to-return signals. Completed is observed from course completion records.

The key measurement challenge is distinguishing Stopped-Out from Withdrawn. The PSF addresses this by treating Intent to Return as a latent variable and using extension requests, tutor contacts, and future registrations as proxies. In the absence of such proxies, the model may remain uncertain, which is itself a useful output. Instead of forcing a binary label, the model reports a probability.

It is important to reiterate that the operational definitions in Section 3.3 are used to generate ground-truth labels for model training and evaluation, not to define the latent states themselves. The latent states are inferred from the model; the operational labels are used to assess whether the model's inferences are valid.

5.3 Censoring and Irregular Sampling

Self-paced data are irregularly sampled and subject to censoring. A student may be observed for 90 days and then leave the observation window while still Stopped-Out. This is right censoring. The survival formulation handles censoring naturally. The HMM formulation handles it by treating the final state as latent and computing the posterior distribution over states at the last observation time.

Missing observations are also common. A student may not log in for a month. The model skips the update step for missing observations and propagates the state forward through the prediction step. This is a standard feature of state-space models and is a major advantage over static models that require complete data.

5.4 Course-Level Versus Program-Level Modeling

The PSF as specified is a **course-level persistence model**. The five states describe a student's relationship to a single course: Registered in the course, Progressing through the course, Stopped-Out from the course, Withdrawn from the

course, or Completed the course. This scope reflects the realities of self-paced learning, where persistence decisions are typically made course by course, and where course-level data are more readily available in institutional systems.

The PSF can be extended to **program-level persistence** by treating courses as nested within programs. In a program-level extension, the latent state would represent the student's relationship to the program as a whole, and course-level states would serve as observations or sub-states. For example, a student who has completed several courses but is currently not registered in any course might be in a program-level Stopped-Out state, even if their most recent course was Completed. A hierarchical model could represent this nesting explicitly, with program-level states at the top and course-level states at the bottom.

We focus on course-level persistence in this paper because it is the level at which most self-paced institutions record data and make intervention decisions. However, we note that program-level persistence is an important extension and discuss it further in Section 9.3.

6. Operational Validation Framework

This section addresses the most important methodological challenge in the paper: how to validate a model of stop-out and attrition when the ground-truth labels are themselves derived from administrative records and observation windows. We propose a five-part validation framework.

6.1 Ground-Truth Definitions

As specified in Section 3.3, we define operational attrition and stop-out independently of the model. The key definitions are:

- **Withdrawn (operational):** Formal withdrawal record, or no return within observation window W (default 24 months) with no active extension or future registration.
- **Stopped-Out (operational):** Inactive for at least T_{pause} (default 60 days), no formal withdrawal, and either returns within W or has an active extension, future registration, or documented intent to return.

These definitions are applied at a fixed classification date. For example, a student's status at month 12 after last activity might be classified as Stopped-Out if they returned by month 24, or Withdrawn if they did not. The model is then trained to predict this classification using data available at month 3, 6, or 9.

6.2 Label Generation Procedure

Labels are generated through the following procedure:

1. **Define the observation window W .** For AU learner-paced courses, use $W = 24$ months as the default, with sensitivity analyses at $W = 12$ and $W = 36$ months.
2. **Define the pause threshold T_{pause} .** Use $T_{\text{pause}} = 60$ days as the default, with sensitivity analyses at 30 and 90 days.
3. **Identify last activity date.** For each student, identify the last date of recorded LMS activity or assignment submission.
4. **Apply operational definitions.** Classify each student as Withdrawn, Stopped-Out, or Completed based on the definitions above.
5. **Create time-stamped training examples.** For each student, create training examples at multiple time points (e.g., 3, 6, 9, and 12 months after last activity) with the corresponding label.
6. **Split data temporally.** Use earlier cohorts for training and later cohorts for testing to avoid temporal leakage.

This procedure ensures that the model is predicting a label that is defined independently of the model's own inferences.

6.3 Expected Timeline for Classification

The classification timeline depends on W and T_{pause} . Under the default settings, a student who becomes inactive is first flagged as potentially Stopped-Out after 60 days. The final classification as Stopped-Out or Withdrawn is determined at 24 months after last activity. For early warning applications, the model predicts the probability of eventual withdrawal at earlier time points (e.g., 3, 6, or 9 months), allowing advisors to intervene before the final classification is known.

Figure 1 presents the state-transition diagram for the PSF.

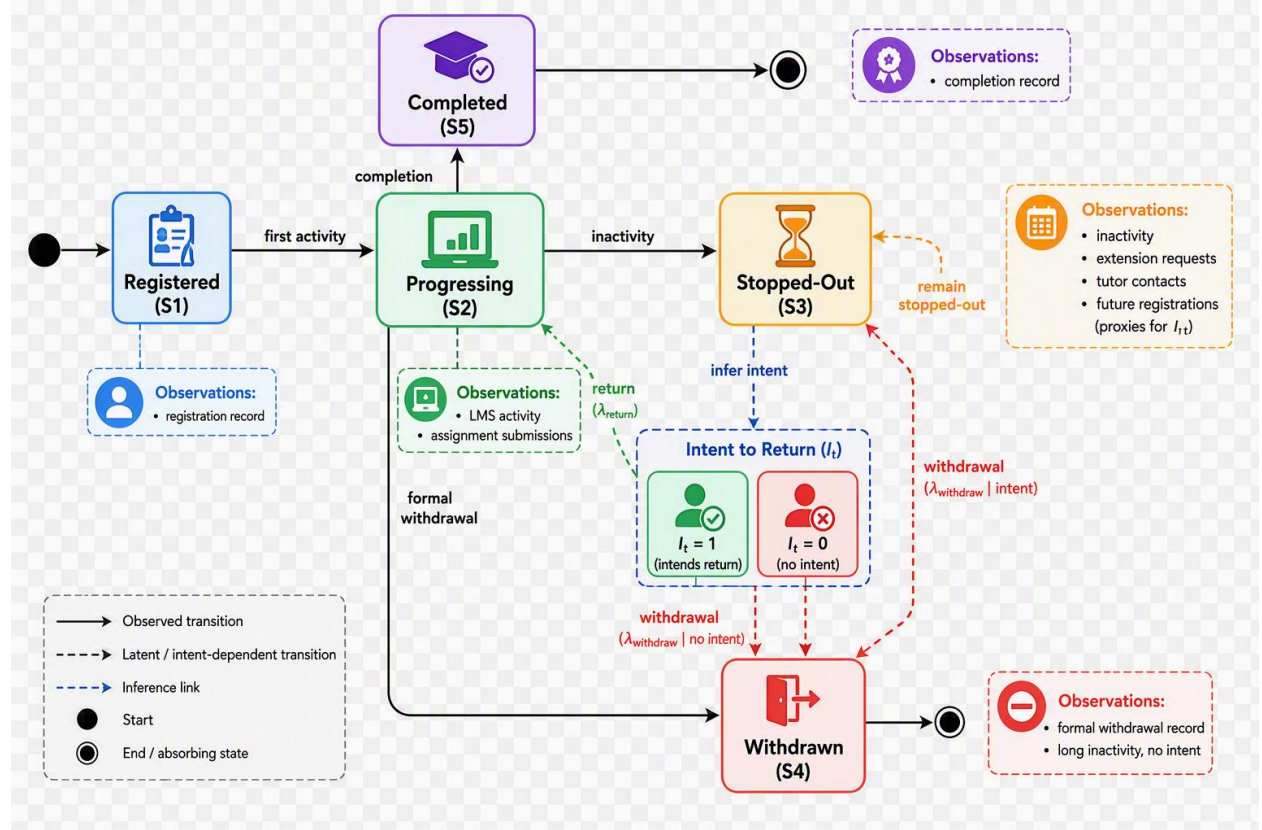


Figure 1. State-transition diagram for the Persistence-State Framework (PSF).

Note on Figure 1. The five persistence states are **Registered (S₁)**, **Progressing (S₂)**, **Stopped-Out (S₃)**, **Withdrawn (S₄)**, and **Completed (S₅)**. Solid edges denote **observed state transitions**, whereas dashed edges denote **latent intent-dependent transitions** mediated by the **Intent-to-Return** latent variable (I_t). The framework's central distinction is between **Stopped-Out (S₃)**, representing temporary disengagement from study, and **Withdrawn (S₄)**, representing permanent attrition.

Transition probabilities originating from **Stopped-Out (S₃)** depend on the inferred value of I_t . The parameter $\lambda_{\text{return}}(t)$ represents the probability of returning to active study when $I_t = 1$. The parameter $\lambda_{\text{withdraw}|\text{intent}}(t)$ represents the probability of withdrawal despite an intention to return. The parameter $\lambda_{\text{withdraw}|\text{no intent}}(t)$ represents the probability of withdrawal when $I_t = 0$.

Observation streams associated with each state are used for inference and validation. **Registered (S_1)** is represented by registration records. **Progressing (S_2)** is represented by learning management system (LMS) activity and assignment submissions. **Stopped-Out (S_3)** is characterized by inactivity periods, extension requests, tutor contacts, and future course registrations, which serve as observable proxies for I_t . **Withdrawn (S_4)** is represented by formal withdrawal records or prolonged inactivity without evidence of intent to return. **Completed (S_5)** is represented by course completion records.

The framework models persistence as a dynamic, state-based process rather than a binary outcome. By explicitly representing latent intent and persistence-state transitions, it provides a computational basis for distinguishing temporary stop-out from permanent attrition in self-paced online learning environments.

6.4 Example Student Trajectories

To illustrate the PSF, consider three hypothetical AU student trajectories.

Trajectory A: Stop-Out with Return.

- Month 0: Registers for a course.
- Month 1: Accesses materials, completes first assignment. (Progressing)
- Month 3: Requests a 2-month extension. (Intent to Return = 1)
- Month 4–6: No LMS activity. (Stopped-Out)
- Month 7: Returns, submits second assignment. (Progressing)
- Month 12: Completes course. (Completed)

Under an inactivity-threshold model, this student would have been flagged as dropped out at Month 4. Under the PSF, the extension request signals Intent to Return, and the model would predict a high probability of return. This is a false positive avoided.

Trajectory B: Stop-Out without Return.

- Month 0: Registers for a course.
- Month 1: Accesses materials, no submissions. (Progressing)
- Month 3: No activity for 60 days. (Stopped-Out)
- Month 5: No extension request, no tutor contact. (Intent to Return = 0)
- Month 24: No return. (Withdrawn)

Under an inactivity-threshold model, this student would be flagged as dropped out at Month 3 or 4. Under the PSF, the absence of intent signals would gradually increase the probability of withdrawal, and the model would correctly classify the student as Withdrawn by Month 12.

Trajectory C: Formal Withdrawal.

- Month 0: Registers for a course.
- Month 1: Accesses materials, completes first assignment. (Progressing)
- Month 2: Formally withdraws. (Withdrawn)

Under both models, this student is correctly classified as Withdrawn. The PSF adds value by distinguishing this case from Trajectory A.

6.5 Illustrative State-Transition Diagram

Figure 1 above presents the state-transition diagram. The diagram shows the five states and the possible transitions among them. The critical transitions are:

- Progressing $\rightarrow$ Stopped-Out (inactivity)
- Stopped-Out $\rightarrow$ Progressing (return, $I_t = 1$)
- Stopped-Out $\rightarrow$ Withdrawn (no return, $I_t = 0$)
- Progressing $\rightarrow$ Completed (completion)
- Progressing $\rightarrow$ Withdrawn (formal withdrawal)

The diagram also shows that Stopped-Out is a transient state: a student cannot remain Stopped-Out indefinitely. Over time, they either return to Progressing or transition to Withdrawn. The rate of transition depends on Intent to Return and other covariates.

7. Evaluation and Comparison Against Traditional Dropout Models

7.1 Baselines

To demonstrate the value of the PSF, we propose comparison against three traditional dropout models.

Baseline 1: Inactivity-threshold classifier. A student is labeled as dropped out if they have no LMS activity for a fixed period, such as 30 days. This is the most common operational definition in practice.

Baseline 2: Static logistic regression. A binary outcome model that predicts dropout at a fixed time point using covariates such as age, prior GPA, registration history, and early activity.

Baseline 3: Survival model without intent. A discrete-time survival model that predicts time to dropout but does not include the Intent to Return variable. This model can distinguish between active and inactive periods but cannot distinguish between temporary stop-out and permanent withdrawal.

The proposed model is the PSF with Intent to Return, implemented as either an HMM/DBN or a competing-risk survival model.

7.2 Metrics

The models are evaluated on their ability to predict withdrawal and return. Appropriate metrics include:

- **Area under the ROC curve (AUC)** for withdrawal prediction.
- **Precision-recall AUC (PR-AUC)** because withdrawal is a relatively rare event.
- **Brier score** for probabilistic calibration.
- **Time-dependent AUC** for survival models.
- **Concordance index** for ranking students by risk.
- **False positive rate for withdrawal alerts**, which is critical for avoiding unnecessary interventions.
- **Return prediction accuracy**, which evaluates the model's ability to identify students who will return.

The central comparison is between the PSF and the inactivity-threshold classifier. The PSF is expected to reduce false positives substantially while maintaining or improving true positive detection of withdrawal.

7.3 Hypotheses

The evaluation tests the following hypotheses:

- H1.** The PSF outperforms the inactivity-threshold classifier on PR-AUC for withdrawal prediction.
- H2.** The PSF reduces the false positive rate for withdrawal alerts compared with the inactivity-threshold classifier.
- H3.** The inclusion of Intent to Return improves return prediction over a survival model without intent.
- H4.** Extension requests and tutor contacts are significant predictors of return, controlling for activity recency.

These hypotheses are tested using historical AU course records, with cross-validation across courses and time periods.

7.4 Sensitivity Analyses

Because the operational definitions of attrition and stop-out depend on W and T_{pause} , we recommend sensitivity analyses across multiple values of these parameters. Specifically:

- $W \in \{12, 24, 36\}$ months.
- $T_{\text{pause}} \in \{30, 60, 90\}$ days.

Results should be reported for each combination, with the default settings ($W = 24$, $T_{\text{pause}} = 60$) as the primary analysis.

8. Applications and Integration with Learning Analytics Systems

8.1 Integration with Learning Analytics Pipelines

The PSF is designed to be integrated into existing learning analytics pipelines and predictive analytics systems. In a typical institutional learning analytics architecture, data flow from the LMS, student information systems, and administrative records into a data warehouse, where they are processed and made available to predictive models. The PSF occupies a specific role in this pipeline: it operates as a **temporal state inference layer** that sits between raw observation streams and downstream decision-support applications.

Unlike static risk scores, which produce a single probability of dropout at a fixed time point, the PSF produces a **time-varying posterior distribution** over persistence states and Intent to Return. This distribution can be updated each time new data arrive — whether from an LMS login, an assignment submission, an extension request, or a tutor contact. The result is a continuously updated belief about whether a student is Progressing, Stopped-Out with intent to return, Stopped-Out without intent, or Withdrawn. This is a natural fit for modern learning analytics systems that increasingly emphasize real-time or near-real-time monitoring.

The PSF Python library supports this integration through its particle filter function, which provides online state estimation as new observations arrive, and through its modular design, which allows the inference layer to be called from existing pipeline orchestration systems without modification. The library's `evaluate_at_positive_rate` function is particularly relevant to pipeline integration: it allows the institution to specify an operational capacity (e.g., "we can contact 100 students per week") and evaluate the model's performance at that capacity directly, rather than at an arbitrary decision threshold that does not reflect operational constraints.

8.2 Early Warning and Intervention Orchestration

The PSF supports early warning systems by computing $\Pr(\text{Withdraw} \mid H_t)$ and $\Pr(\text{Return} \mid H_t)$ for each student at each time step. These probabilities can be fed into intervention orchestration systems — platforms that automate the selection, timing, and delivery of support actions. For example, a student with a high probability of return and a moderate probability of withdrawal might receive a light-touch reminder or an extension option. A student with a low probability of return and a high probability of withdrawal might be escalated to a human advisor.

This integration is particularly valuable because it allows interventions to be **tiered by urgency and likelihood of response**. Traditional dropout models produce a binary flag (at risk / not at risk) that treats all flagged students identically. The PSF produces a richer set of outputs — probability of return, probability of withdrawal, expected time to return, and uncertainty about Intent to Return — that allow institutions to allocate scarce support resources more efficiently and to avoid over-intervening with students who are likely to return on their own.

The PSF library operationalizes this tiering through its `return_probabilities` and `withdrawal_probabilities` functions, which compute horizon-specific probabilities for any desired horizon (e.g., "will this student return within the next 3 months?"). The library also provides per-student state trajectories and cumulative withdrawal curves (see the `plot_state_trajectory` and `plot_withdrawal_curves` functions) that advisors can inspect to understand *why* a particular student is flagged. This explanatory capability — the ability to move from a probability to an interpretable trajectory — is a distinctive feature of the PSF relative to black-box predictive models, and it aligns with the growing emphasis in learning analytics on transparent and actionable models.

The reference simulation described in Section 4.5.3 illustrates the operational advantage. At matched capacity, the PSF flags 236 true withdrawals out of 239 flagged students (precision 0.987), compared to 214 for the inactivity threshold (precision 0.895). This translates to 22 fewer unnecessary interventions per 239 contact slots — a reduction of more than 40% in wasted outreach, without any reduction in the number of true withdrawals reached.

8.3 Program Evaluation and Policy Reporting

Aggregating PSF outputs across cohorts identifies bottleneck courses where many students stop out and do not return. It also allows institutions to compare stop-out and return rates across programs, courses, and student populations. This is particularly valuable for open and distance universities that serve adult learners with diverse life circumstances.

Traditional completion rates treat all non-completers as attrition. The PSF provides a more nuanced picture: some non-completers are Stopped-Out and may return; others are Withdrawn; others may complete later. This distinction is important for institutional reporting, accreditation, and policy development.

8.4 Personalized Pacing and Support

Estimated state probabilities and return probabilities support recommendations for optimal pacing—slowing down if risk is rising, accelerating if progress is steady—and for adjusting support intensity based on Intent to Return. The PSF can be integrated with recommender systems to deliver personalized content, pacing, and support recommendations in real time.

9. Discussion

9.1 Contributions

The paper makes four primary contributions. First, it develops a formal theory of stop-out as a temporary state with an associated probability of return. Second, it introduces the latent Intent to Return variable and specifies how it can be inferred from administrative and interaction data, with explicit attention to construct, measurement, and proxy validity. Third, it provides computational specifications using HMMs, DBNs, survival analysis, and event history analysis. Fourth, it proposes an operational validation framework that includes ground-truth definitions, label-generation procedures, example trajectories, and a comparison against traditional dropout models. In addition to these four theoretical contributions, the paper presents an **open-source reference implementation** — the PSF Python library — that operationalizes the framework and provides the fitting, inference, validation, and visualization tools needed for empirical work on institutional data. The library is itself a contribution because it reduces the barrier between the theoretical framework and its practical application, and because it enables reproducible validation of the framework's core claims.

9.2 Limitations

Several limitations warrant acknowledgment. First, this is a conceptual and computational framework paper. It does not report empirical model fitting, parameter estimation, or validation against institutional data. The PSF is designed to be implemented and tested in future empirical work. Second, the model's complexity introduces computational demands, particularly for large-scale deployment. Variational inference or distributed particle filtering may be necessary. Third, parameter estimation requires substantial historical data, which may not be available in new programs. Fourth, the Intent to Return variable is difficult to observe directly and must be inferred from proxies. As discussed in Section 3.5, this raises questions of construct, measurement, and proxy validity that must be addressed empirically. Fifth, the model assumes that offline learning can be partially captured through periodic assessments or self-reports. In fully self-paced contexts without such checkpoints, offline engagement may remain entirely latent. Sixth, the choice of observation window W and pause threshold T_{pause} affects the operational definitions of stop-out and attrition, and results may vary across different values. Seventh, the reference implementation is research-grade software; it is intended to support empirical validation, not turnkey deployment. Institutions that adopt the PSF should plan for local validation, calibration monitoring, and institutional review before using the model's outputs to make consequential decisions. Eighth, the model's ethical deployment requires attention to privacy, transparency, and bias. Students should know how persistence is measured, and interventions should support rather than surveil.

9.3 Future Directions

Future work should validate the PSF empirically using data from self-paced institutions such as Athabasca University. This includes estimating model parameters, testing the four hypotheses specified in Section 7.3, and conducting sensitivity analyses across different values of W and T_{pause} . The PSF Python library provides the tooling needed for this empirical work and includes templates for loading institutional data, fitting the model with multi-restart EM, evaluating against the three baselines, and generating publication-ready diagnostic figures. Deep extensions replacing linear transition and observation models with recurrent neural networks could capture complex nonlinear dynamics while retaining interpretable state structure; the library's modular design allows these extensions to be implemented without modifying the core inference algorithms. Multi-level extensions incorporating course-level and program-level random effects would enable more nuanced institutional analysis, as discussed in Section 5.4. Finally, the PSF could be integrated with recommender systems to deliver personalized pacing and support recommendations in real time..

10. Conclusion

This paper has addressed the question: Can stop-out and attrition be distinguished computationally? The answer is yes, provided that the model represents persistence states, latent Intent to Return, and the temporal history of administrative and interaction data. The proposed **Persistence-State Framework (PSF)** separates Registered, Progressing, Stopped-Out, Withdrawn, and Completed states, and formalizes $\Pr(\text{Return} \mid \text{History})$ as a competing-risk problem. Computational specifications using HMMs, DBNs, survival analysis, and event history analysis make the PSF testable and practical. The operational validation framework addresses the ground-truth problem by defining attrition and stop-out independently of the model and specifying label-generation procedures. By distinguishing temporary stop-out from permanent attrition, the PSF provides a foundation for early warning, intervention triage, program evaluation, and policy reporting. It is particularly suited to institutions like Athabasca University, where the distinction between Stopped-Out and Withdrawn is not merely an academic nuance but a practical necessity for supporting adult learners. As self-paced and distance education continues to expand, computational models that respect the complexity of persistence will become increasingly essential for promoting student success.

References

- Athabasca University. (2003). *Completion rates for learner-paced undergraduate courses*. Athabasca University.
- Bean, J. P. (1980). Dropouts and turnover: The synthesis and test of a causal model of student attrition. *Research in Higher Education*, 12(2), 155–187. <https://doi.org/10.1007/BF00976194>

- Bean, J. P., & Metzner, B. S. (1985). A conceptual model of nontraditional undergraduate student attrition. *Review of Educational Research*, 55(4), 485–540. <https://doi.org/10.3102/00346543055004485>
- Chen, F., & Cui, Y. (2020). Utilizing student time series behaviour in learning management systems for early prediction of course performance. *Journal of Learning Analytics*, 7(2), 1–17. <https://doi.org/10.18608/jla.2020.72.1>
- Cocca, M., & Weibelzahl, S. (2011). Student engagement modeling using Bayesian networks. In *Proceedings of the IEEE International Conference on Systems, Man, and Cybernetics* (pp. 2936–2941). IEEE. <https://doi.org/10.1109/ICSMC.2011.6084097>
- Cox, D. R. (1972). Regression models and life-tables. *Journal of the Royal Statistical Society: Series B*, 34(2), 187–220. <https://doi.org/10.1111/j.2517-6161.1972.tb00899.x>
- Dron, J. (2007). *Self-paced and social*. Athabasca University.
- Gašević, D., Dawson, S., Rogers, T., & Gasevic, D. (2016). Learning analytics should not promote one size fits all: The effects of instructional conditions in predicting academic success. *The Internet and Higher Education*, 28, 68–84. <https://doi.org/10.1016/j.iheduc.2015.10.002>
- Hart, C. (2012). Factors associated with student persistence in an online program of study: A review of the literature. *Journal of Interactive Online Learning*, 11(1), 19–42.
- Henrie, C. R., Halverson, L. R., & Graham, C. R. (2015). Measuring student engagement in technology-mediated learning: A review. *Computers & Education*, 90, 36–53. <https://doi.org/10.1016/j.compedu.2015.09.005>
- Johnson, I. Y. (2006). *Analysis of stopout behavior at a public research university: The multi-spell discrete-time approach* (Doctoral dissertation, Ohio State University).
- Jovanović, J., Gašević, D., Dawson, S., Pardo, A., & Mirriahi, N. (2017). Learning analytics to unveil learning strategies in a flipped classroom. *The Internet and Higher Education*, 33, 74–85. <https://doi.org/10.1016/j.iheduc.2017.02.001>
- Kember, D. (1995). *Open learning courses for adults: A model of student progress*. Educational Technology Publications.
- Koller, D., & Friedman, N. (2009). *Probabilistic graphical models: Principles and techniques*. MIT Press.
- Matcha, W., Uzir, N. A., Gašević, D., & Pardo, A. (2020). A systematic review of empirical studies on learning analytics dashboards: A self-regulated learning perspective. *IEEE Transactions on Learning Technologies*, 13(2), 226–245. <https://doi.org/10.1109/TLT.2019.2916802>
- Murphy, K. P. (2002). *Dynamic Bayesian networks: Representation, inference and learning* (Doctoral dissertation, University of California, Berkeley).

- Park, J. H., & Choi, H. J. (2009). Factors influencing adult learners' decision to drop out or persist in online learning. *Educational Technology & Society*, 12(4), 207–217.
- Rabiner, L. R. (1989). A tutorial on hidden Markov models and selected applications in speech recognition. *Proceedings of the IEEE*, 77(2), 257–286. <https://doi.org/10.1109/5.18626>
- Rovai, A. P. (2003). In search of higher persistence rates in distance education online programs. *The Internet and Higher Education*, 6(1), 1–16. [https://doi.org/10.1016/S1096-7516\(02\)00158-6](https://doi.org/10.1016/S1096-7516(02)00158-6)
- Shefsky, E., & Sutton, R. (2015). Stop-outs: A new variable in online student retention. In D. Rutledge & D. Slykhuis (Eds.), *Proceedings of SITE 2015: Society for Information Technology & Teacher Education International Conference* (pp. 480–487). Association for the Advancement of Computing in Education.
- Siemens, G., & Long, P. (2011). Penetrating the fog: Analytics in learning and education. *EDUCAUSE Review*, 46(5), 30–32.
- Simpson, O. (2013). Student retention in distance education: Are we failing our students? *Open Learning: The Journal of Open, Distance and e-Learning*, 28(2), 105–119. <https://doi.org/10.1080/02680513.2013.847363>
- Subramaniam, T. T., Ismail, M. R., Malik, M. A., & Hamidon, Z. (2024). Exploring learner autonomy and self-regulated learning in open-distance education: Impacts on engagement, performance, and instructional strategies. *ASEAN Journal of Open and Distance Learning*, 16(1).
- Tinto, V. (1975). Dropout from higher education: A theoretical synthesis of recent research. *Review of Educational Research*, 45(1), 89–125. <https://doi.org/10.3102/00346543045001089>
- Tinto, V. (1993). *Leaving college: Rethinking the causes and cures of student attrition* (2nd ed.). University of Chicago Press.
- Wang, Y., & Han, X. (2016). Inference mechanism in Bayesian networks for adaptive learning environments. *Journal of Educational Data Mining*, 8(1), 1–22.
- Xavier, M., Meneses, J., & Fiuza, P. J. (2026). Dropout, stopout, and time challenges in open online higher education: A qualitative study of the first-year student experience. *Open Learning: The Journal of Open, Distance and e-Learning*, 41(1), 24–40.
- Zhang, Y., et al. (2024). Memory-based dynamic Bayesian networks for learner modeling: Towards early prediction of learners' performance in computational thinking. *Applied Sciences*, 14(16), 7123. <https://doi.org/10.3390/app14167123>
- Zimmerman, B. J. (2002). Becoming a self-regulated learner: An overview. *Theory Into Practice*, 41(2), 64–70. https://doi.org/10.1207/S15430421TIP4102_2